Slot - C
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(1) Computer vision:-
Computer vision enable Computers to understand the video's and images. It uses images processing and AI Techniques to identify objects and patterns.

(1) a] The main objectives of Computer vision are image anaylis and object identification, decision-making by interpreting visual information. It Increases the effiency and accuracy.

(1) b] Applications:- i] Face re detection. Systemes.
ii] Attendance syestem using face detection
iii] self-Driving cars detect roads and obstacles.

(2) a] Feature extraction:-
The key features of Computer vision identifying the images and videos and important Characteristics such as edges, Corners, and textures. These features are useful to detect the images

b] object recognition identifies and classifies objects present in an image.

③ i] Low-Level image processing.
ii] mid-Level image processing.
iii] high-Level image processing.

i] Low Level image processing:-
It includes filltering, noise, removal, and enhancement output in another is improved.

ii] mid-Level
*) mid-Level extracts features and segments objects from images.
*) deals with image analysis

high-Level
*) It processing interprets the extracted information to recognize objects.
*) deals with understanding and decision-making

④
a] Pixels:-
A pixel is the smallest element of a digital image. Each pixel stores brightness or color information. millions of pixels together form an image. pixel values determine image quality.

b] Image resolution refers to the number of pixels in an image. Higher resolution provides more detail and clarity and low resolution causes to blurred images.

An image sensor can converts image into light signals and the signals. They are processed into digital images. It is the main component of a camera. it plays a main role in image (or) video processing.

Ex:- i] digital cameras

ii] scanner's

5)

a] Sampling converts a continuous image into a grid of pixels. it determines the spatial resolution of the image. more samples provide better detail. it is the first step in digital image formation.

b] Quantization assigns numerical intensity values to sampled pixels. it converts continuous brightness into discrete levels. This ~~allows~~ the images may stores digitally.

7)

a] The perspective of projection converts a 3D sence into a 2D image. it creates a realistic vew of objects. Distant objects appear smaller than nearby ones. it is main to used in cameras.

b] Illumination provides the light required to capure images. proper lighting improves image quality. poor illumination reduces visibility and accuracy.

(8)

a] Image processing enhances or modifies images using algorithms. it improves image quality using removing noise (or) adjusting brightness.

b] Image processing focuses on improving images while Computer vision focuses on understanding image content. Image processing produces enhanced images.

(9)

a] In healthcare, Computer vision helps detect diseases from x-rays, CT scans, and MRI images. It assists doctors in diagnosis. It improves accuracy and reduces manual effort.

b] In autonomous vehicles, Computer vision detects roads, traffic signs, pedestrians, and obstacles. it helps vehicles navigate safely. It enables lane detection and object tracking it improves road safety.

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(10) a] pixels are the basic building blocks of digital images. Each pixel stores color or intensity information Together they form the complete image. more pixels result in better image quality.

b] Sampling determines the number of pixels captured while resolution determines image detail. Higher sampling and resolution improve analysis accuracy.